TARGET SHIP DATE: 2026-04-22
Gerrymandering Simulator by shiii_man
Roster: Isabel Zheng (PM), Haowen Xiao, Sean Takahashi, Araf Hoque, and Owen Zeng

Overview:

Using the New York Open Data files on voter registration by NYC election districts and the median salaries by congressional district, we can identify whether district boundaries divide populations in ways consistent with gerrymandering (ex. Packing wealthy voters together). We intend for this to be interactive, illustrating a map of New York (colors indicate varying levels of income & drawn county and congressional district lines) and allowing users to learn how gerrymandering is done. This was our inspiration for this project in terms of detailed voting results: https://votehub.com/2024-map/

Database Organization:

accounts		
INTEGER	id	PK AUTOINCREMENT
TEXT	username	UNIQUE
TEXT	password	

All gerrymandered maps

saved_maps		
INTEGER	id	PK AUTOINCREMENT
INTEGER	user_id	
TEXT	map_name	
TIMESTAMP	created_at	CURRENT_TIMESTAMP

Singular gerrymandered map

map_gmed		
INTEGER	id	PK AUTOINCREMENT
INTEGER	map_id	UNIQUE
INTEGER	election_district	

INTEGER	assembly_district	
INTEGER	assigned_district_id	

Core Components:

1. CSV Dataset Files
    a. Both datasets are relational. To observe the potential gerrymandering techniques, the data needs to be stored in rows and columns; the overlaps between voting districts and ZIP codes in terms of their geographical area.
        i. Voter Data 2008-2018 (https://data.cityofnewyork.us/City-Government/Voter-Analysis-2008-2018/psx2-aqx3/about_data): Gives data of NYC voters who have voted between 2008 and 2018 including their party registration, what elections they voted in, and their geographic location(voting district)
        ii. Salary Information 2018-Adjusted (https://data.census.gov/all?q=median+salary&g=040XX00US36$8600000): Taken from Census data, gives salary information including median, mean, quartile of yearly income by ZIP code.
2. GeoJSON files
    a. ny_election_districts.geojson: Each election district stored as their geographic area
    b. ny_zipcodes.geojson: NYC based on ZIP code.
3. Geopandas
    a. Links counties & districts (spatial joins for later analysis)
4. Python
    a. __init__.py: Serve the Flask app, base file
    b. data.py: Handles unpacking of CSV data, extracting necessary information
    c. auth.py: Basic login/register functionality
5. Folium for Visualization (creates an interactive map, generating JS and HTML) → has to give location points

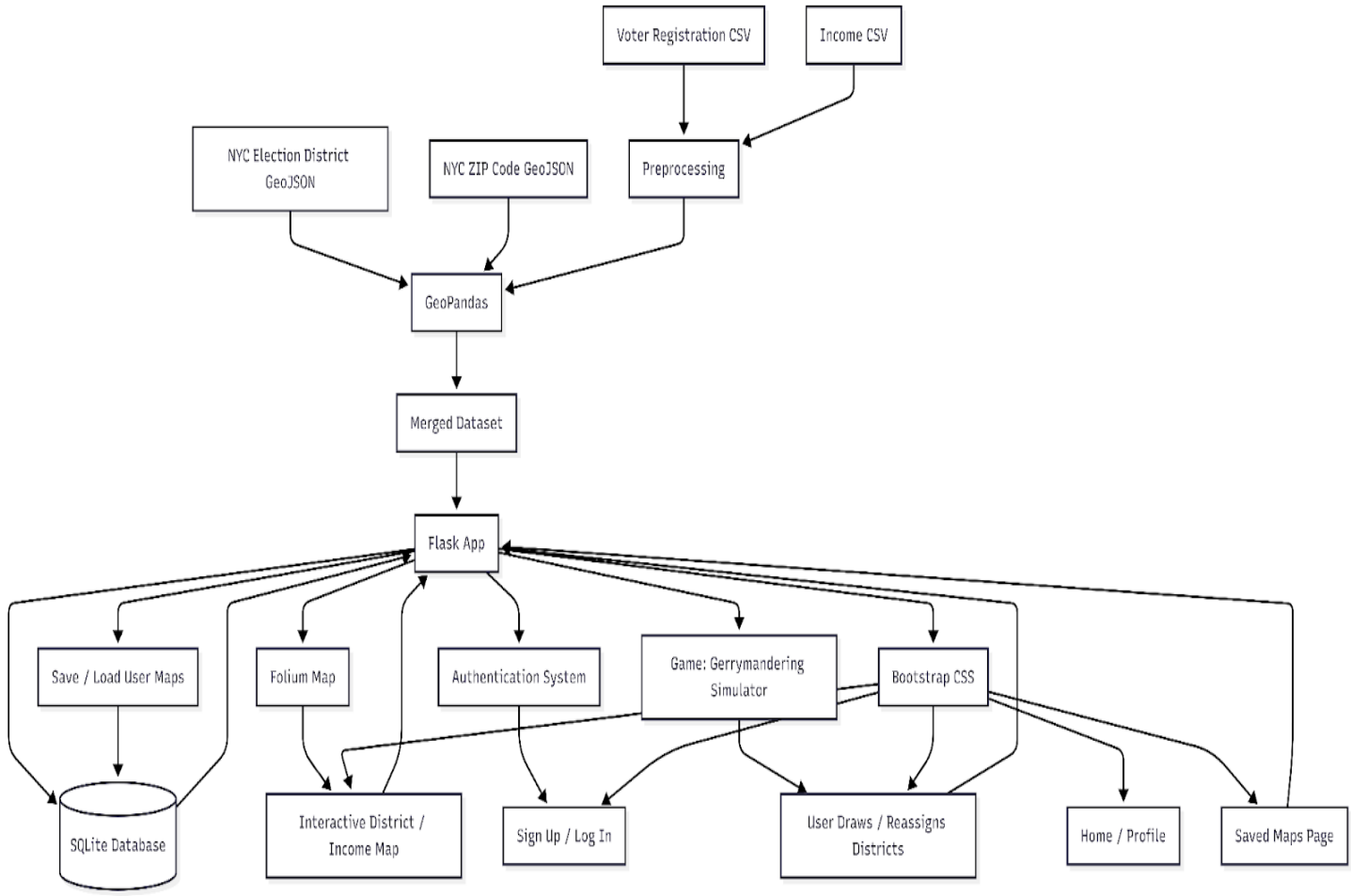
Front-end Framework: Bootstrap (contains prebuilt CSS classes for buttons, containers, grid layouts. These prebuilt classes make styling much easier and more time efficient, allowing us to focus more on the backend: game logic, data processing, etc)

APIs: We don't intend on using any APIs for this project.

Data Visualization Library:

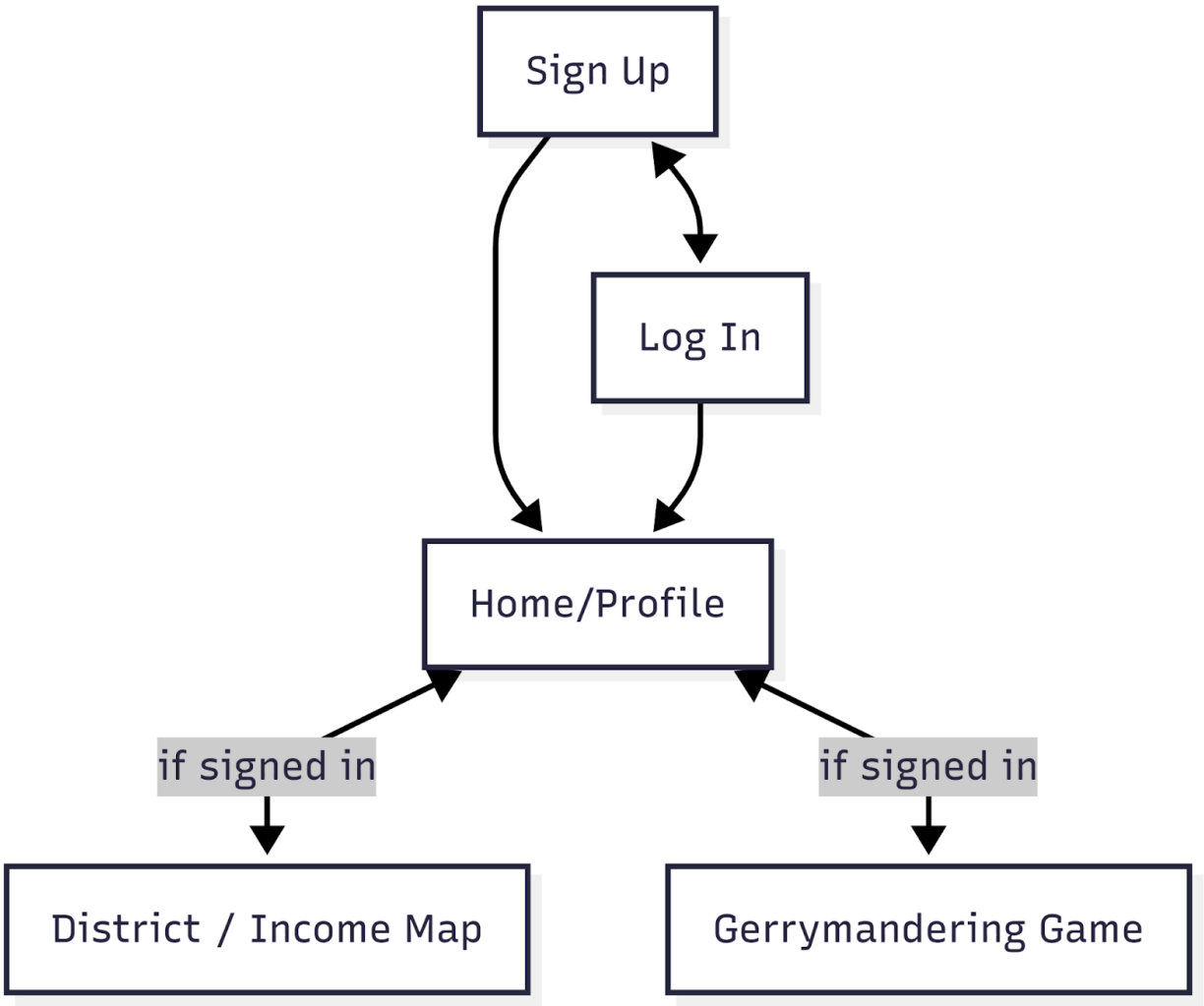
- Folium: We need to use a library that allows us to interact with a map & Folium takes spatial and relational data to create this. It will show the election districts, ZIP code, income distribution, and voter distribution with colors and lines as explained previously.

Component Map:



[(Mermaid Live Editor)]

Site Map:



(Mermaid Live Editor)

https://gist.github.com/abatko/ee7b24db82a6f50cfce02afafa1dfd1e